

Online Retail Order Processing System using Apache Kafka

1. Introduction

This document presents the design and implementation of an online retail order processing system using Apache Kafka CLI. The system demonstrates Kafka concepts such as producers, consumers, topics, partitions, consumer groups, replication, and message key-based partitioning.

Scenario:

Customers place orders that are processed by multiple services:

- Inventory Service: Updates stock
- Billing Service: Generates invoices
- Shipping Service: Prepares shipments

2. Objectives

- Explore Kafka producers and consumers
- Create and manage Kafka topics and partitions
- Simulate consumer groups and load balancing
- Understand replication factor and fault tolerance
- Control partition assignment using message keys

3. System Architecture

Customer Orders --> Kafka Orders Topic --> Consumers:

Inventory Service
Billing Service
Shipping Service

Topic: orders

Partitions: 3

Replication Factor: 2

Consumer Group: retailGroup

4. Implementation Steps

Step 1: Start Kafka and Zookeeper

```
zookeeper-server-start.sh config/zookeeper.properties
```

```
kafka-server-start.sh config/server.properties
```

```

...epeer.properties ...server.properties ...server2.properties ...server3.properties ...-r localhost:9092 ...print.offset=true ...-r localhost:9092 ...-5./iconfig--zsh ...-u mysqladmin-p +
[2025-11-12 16:37:19,051] INFO Using checkIntervalMs=60000 maxPerMinute=10000 maxNeverUsedIntervalMs=0 (org.apache.zookeeper.server.ContainerManager)
[2025-11-12 16:37:19,052] INFO ZooKeeper audit is disabled. (org.apache.zookeeper.audit.ZAuditProvider)
[2025-11-12 16:39:12,026] INFO Creating new log file: log-1 (org.apache.zookeeper.server.persistence.FileTxnLog)
+
[2025-11-12 16:39:12,033] config # zookeeper-server-start.sh SKAFKA_PATH/config/zookeeper.properties
[2025-11-12 17:03:57,657] INFO Reading configuration from: /Users/racit/Downloads/kafka_2.12-3.5.1/bin/config/zookeeper.properties (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-11-12 17:03:57,657] INFO ClientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-11-12 17:03:57,657] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-11-12 17:03:57,657] INFO observerMasterPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-11-12 17:03:57,657] INFO metricsProvider.className is org.apache.zookeeper.metrics.impl.DefaultMetricsProvider (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-11-12 17:03:57,658] INFO autoPurge.snapRetainCount set to 3 (org.apache.zookeeper.server.DataDirCleanupManager)
[2025-11-12 17:03:57,658] INFO autoPurge.purgeInterval set to 0 (org.apache.zookeeper.server.DataDirCleanupManager)
[2025-11-12 17:03:57,658] INFO Purge task is not scheduled. (org.apache.zookeeper.server.DataDirCleanupManager)
[2025-11-12 17:03:57,658] WARN Either no config or no quorum defined in config, running in standalone mode (org.apache.zookeeper.server.quorum.QuorumPeerMain)
[2025-11-12 17:03:57,659] INFO Log4j 1.2 jmx support not found; jmx disabled. (org.apache.zookeeper.jmx.ManagedUtil)
[2025-11-12 17:03:57,659] INFO Reading configuration from: /Users/racit/Downloads/kafka_2.12-3.5.1/bin/config/zookeeper.properties (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-11-12 17:03:57,659] INFO ClientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-11-12 17:03:57,659] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-11-12 17:03:57,659] INFO observerMasterPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-11-12 17:03:57,659] INFO metricsProvider.className is org.apache.zookeeper.metrics.impl.DefaultMetricsProvider (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-11-12 17:03:57,660] INFO Starting server (org.apache.zookeeper.server.ZooKeeperServerMain)
[2025-11-12 17:03:57,663] INFO ServerMetrics initialized with provider org.apache.zookeeper.metrics.impl.DefaultMetricsProvider@29f69900 (org.apache.zookeeper.server.ServerMetrics)
[2025-11-12 17:03:57,664] INFO zookeeper.snapshot.trust.empty = false (org.apache.zookeeper.server.persistence.FileTxnSnapLog)
[2025-11-12 17:03:57,667] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-11-12 17:03:57,667] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-11-12 17:03:57,667] INFO (org.apache.zookeeper.server.ZooKeeperServer)
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[2025-11-12 17:03:57,668] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-11-12 17:03:57,668] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-11-12 17:03:57,671] INFO Server environment:zookeeper.version=3.6.4--65623dc68e907c6e9a1b26453f5f5Feb452ic, built on 12/18/2022 18:19 GMT (org.apache.zookeeper.server.ZooKeeperServer)
[2025-11-12 17:03:57,671] INFO Server environment:host.name=72.16.26.142 (org.apache.zookeeper.server.ZooKeeperServer)
[2025-11-12 17:03:57,671] INFO Server environment:java.version=7.0.12 (org.apache.zookeeper.server.ZooKeeperServer)
[2025-11-12 17:03:57,671] INFO Server environment:java.vendor=Oracle Corporation (org.apache.zookeeper.server.ZooKeeperServer)
[2025-11-12 17:03:57,671] INFO Server environment:java.home=/Library/Java/JavaVirtualMachines/jdk-7.jdk/Contents/Home (org.apache.zookeeper.server.ZooKeeperServer)
[2025-11-12 17:03:57,671] INFO Server environment:java.class.path=/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/activation-1.1.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/apollance-repackaged-2.6.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/argonauts-2.7.0.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/audience-annotations-0.13.0.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/commons-cli-1.4.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/commons-lang3-3.8.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/connect-api-3.5.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/connect-basic-auth-extension-3.5.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/connect-jmx-3.5.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/connect-mirror-3.5.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/connect-runtime-3.5.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/conne-transforms-3.5.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/HK2-api-2.6.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/HK2-locator-2.6.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/HK2-utilities-2.6.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jackson-annotations-2.13.5.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jackson-core-2.13.5.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jackson-databind-2.13.5.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jackson-dataformat-csv-2.13.5.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jackson-datatype-jdk8-2.13.5.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jackson-jaxrs-base-2.13.5.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jackson-jaxrs-json-provider-2.13.5.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jakarta.annotation-api-1.3.5.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jakarta.inject-2.6.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jakarta.validation-api-2.0.2.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jakarta.ws.rs-api-2.1.6.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jakarta.xml.bind-api-2.3.3.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/javassist-3.29.2-GA.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/javax.activation-api-1.2.0.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/javax.annotation-api-1.3.2.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/javax.servlet-api-3.1.0.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/javax.ws.rs-api-2.1.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jaxb-api-2.3.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jersey-client-2.39.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jersey-common-2.39.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jersey-container-servlet-2.39.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jersey-container-servlet-core-2.39.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jersey-hk2-2.39.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jersey-media-jaxb-2.39.1.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jetty-9.4.51.v20238217.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jetty-http-9.4.51.v20238217.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jetty-io-9.4.51.v20238217.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jetty-security-9.4.51.v20238217.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jetty-server-9.4.51.v20238217.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jetty-servlet-9.4.51.v20238217.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jetty-xml-9.4.51.v20238217.jar:/Users/racit/Downloads/kafka_2.12-3.5.1/bin/./libs/jline-3.22.6.jar:/Users/racit/D
[2025-11-12 16:40:40,807] INFO Registered kafka:type=kafka.Log4jController MBean (kafka.utils.Log4jControllerRegistration)
[2025-11-12 16:40:40,125] INFO Setting -r jdk.tls.rejectClientInitiatedRenegotiation=true to disable client-initiated TLS renegotiation (org.apache.zookeeper.common.XS90Util)
[2025-11-12 16:40:40,149] INFO Registered signal handlers for INT, HUP (org.apache.kafka.common.utils.LoggingSignalHandler)
[2025-11-12 16:40:40,150] INFO starting (kafka.server.KafkaServer)
[2025-11-12 16:40:40,150] INFO Connecting to zookeeper on localhost:2181 (kafka.server.KafkaServer)
[2025-11-12 16:40:40,158] INFO [ZooKeeperClient Kafka server] Initializing a new session to localhost:2181. (kafka.zookeeper.ZooKeeperClient)
[2025-11-12 16:40:40,193] INFO Client environment:zookeeper.version=3.6.4--65623dc68e907c6e9a1b26453f5f5Feb452ic, built on 12/18/2022 18:19 GMT (org.apache.zookeeper.ZooKeeper)
[2025-11-12 16:40:40,193] INFO Client environment:host.name=72.16.26.142 (org.apache.zookeeper.ZooKeeper)
[2025-11-12 16:40:40,193] INFO Client environment:java.version=7.0.12 (org.apache.zookeeper.ZooKeeper)
[2025-11-1
```

Step 2: Create Kafka Topics

```
kafka-topics.sh --create --topic orders --bootstrap-server localhost:9092 --partitions 3  
--replication-factor 2  
kafka-topics.sh --list --bootstrap-server localhost:9092  
kafka-topics.sh --describe --topic orders --bootstrap-server localhost:9092
```

```
racit@PTPLMRT03 ~ % kafka-topics.sh --create --topic orders --bootstrap-server localhost:9092 --partitions 3 --replication-factor 2  
Created topic orders.  
racit@PTPLMRT03 ~ % kafka-topics.sh --bootstrap-server localhost:9092 --list  
__consumer_offsets  
firsttopic  
orders  
racit@PTPLMRT03 ~ % kafka-topics.sh --describe --topic orders --bootstrap-server localhost:9092  
Topic: orders   TopicId: P5bIYNRpQUuAE61q9S37pw PartitionCount: 3      ReplicationFactor: 2   Configs:  
    Topic: orders Partition: 0   Leader: 0      Replicas: 0,1   Isr: 0,1  
    Topic: orders Partition: 1   Leader: 2      Replicas: 2,0   Isr: 2,0  
    Topic: orders Partition: 2   Leader: 1      Replicas: 1,2   Isr: 1,2
```

Step 3: Produce Messages

```
kafka-console-producer.sh --topic orders --bootstrap-server localhost:9092  
Sample messages:  
OrderID:1001,Item:Laptop,Quantity:2  
OrderID:1002,Item:Mobile,Quantity:1  
OrderID:1003,Item:Headphones,Quantity:5
```

```
racit@PTPLMRT03 ~ % kafka-console-producer.sh --topic orders --bootstrap-server localhost:9092  
>OrderID:1001,Item:Laptop,Quantity:2  
>OrderID:1002,Item:Mobile,Quantity:1  
>OrderID:1003,Item:Headphones,Quantity:5  
>
```

Step 4: Consume Messages

```
kafka-console-consumer.sh --topic orders --from-beginning --bootstrap-server  
localhost:9092 --property print.partition=true --property print.offset=true  
Observation: Messages are distributed across partitions.  
Format: partition:offset:message
```

```
racit@PTPLMRT03 ~ % kafka-console-consumer.sh --topic orders --from-beginning --bootstrap-server localhost:9092 --property print.partition=true --property print.offset=true  
Partition:1   Offset:0      OrderID:1001,Item:Laptop,Quantity:2  
Partition:1   Offset:1      OrderID:1002,Item:Mobile,Quantity:1  
Partition:1   Offset:2      OrderID:1003,Item:Headphones,Quantity:5  
  
Partition:0   Offset:0      Laptop  
Partition:0   Offset:1      Mobile  
Partition:0   Offset:2      Mouse
```

Step 5: Use Consumer Groups

```
kafka-console-consumer.sh --topic orders --group retailGroup --bootstrap-server localhost:9092
```

```
kafka-consumer-groups.sh --bootstrap-server localhost:9092 --list
```

```
kafka-consumer-groups.sh --bootstrap-server localhost:9092 --describe --group retailGroup
```

Observation: Consumers share partitions, achieving load balancing.

```
racit@PTPLMRT03 config % kafka-console-consumer.sh --topic orders --group retailGroup --bootstrap-server localhost:9092
^CProcessed a total of 0 messages
racit@PTPLMRT03 config % kafka-console-consumer.sh --topic orders --group retailGroup --bootstrap-server localhost:9092
kafka-console-consumer.sh --topic orders --group retailGroup --bootstrap-server localhost:9092

Laptop
Mobile
Mouse
```

```
racit@192 config % kafka-consumer-groups.sh --bootstrap-server localhost:9092 --list
retailGroup
console-consumer-45075
racit@192 config % kafka-consumer-groups.sh --bootstrap-server localhost:9092 --describe --group retailGroup
```

GROUP	TOPIC	PARTITION	CURRENT-OFFSET	LOG-END-OFFSET	LAG	CONSUMER-ID	HOST	CLIENT-ID
retailGroup	orders	0	0	0	0	console-consumer-33cf36e7-2670-43af-b6fe-a85dd70ba63a	/127.0.0.1	console-consumer
retailGroup	orders	1	3	3	0	console-consumer-33cf36e7-2670-43af-b6fe-a85dd70ba63a	/127.0.0.1	console-consumer
retailGroup	orders	2	0	0	0	console-consumer-33cf36e7-2670-43af-b6fe-a85dd70ba63a	/127.0.0.1	console-consumer

```
racit@192 config % kafka-console-producer.sh \
--topic orders \
--property parse.key=true \
--property key.separator=: \
--bootstrap-server localhost:9092
```

Step 6: Produce Messages with Keys

```
kafka-console-producer.sh --topic orders --bootstrap-server localhost:9092 --property parse.key=true --property key.separator=:
```

Sample keyed messages:

1001:Laptop

1002:Mobile

1001:Mouse

Observation: Messages with the same key go to the same partition.

```
racit@192 config % kafka-console-producer.sh --topic orders --bootstrap-server localhost:9092 --property parse.key=true --property key.separator=:
>1001:Laptop
1002:Mobile
1001:Mouse
```

Step 7: Explore Replication

```
kafka-topics.sh --describe --topic orders --bootstrap-server localhost:9092
```

Observation: Leaders and replicas provide fault tolerance.

Messages remain available even if a broker stops.

Step 8: Cleanup

```
kafka-topics.sh --delete --topic orders --bootstrap-server localhost:9092
```